

# CBS 2372

# CRYPTOGRAPHY

AES Simulation CrypTool2

Lab Work


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# About

Tool used : **CrypTool 2**

CrypTool 2 (CT2) is a modern e-learning program for Windows, which visualizes cryptography and cryptanalysis. It includes not only the encryption and cryptanalysis of ciphers, but also their basics and the whole spectrum of modern cryptography.

Encryption used : **Advanced Encryption Standard (AES)**

Advanced Encryption Standard (AES) is a highly trusted encryption algorithm used to secure data by converting it into an unreadable format without the proper key. It was developed by the National Institute of Standards and Technology (NIST) in 2001. It is widely used today as it is much stronger than DES and triple DES despite being harder to implement. AES encryption uses various key lengths (128, 192, or 256 bits) to provide strong protection against unauthorized access. This data security measure is efficient and widely implemented in securing internet communication, protecting sensitive data, and encrypting files. AES, a cornerstone of modern cryptography, is recognized globally for its ability to keep information safe from cyber threats.



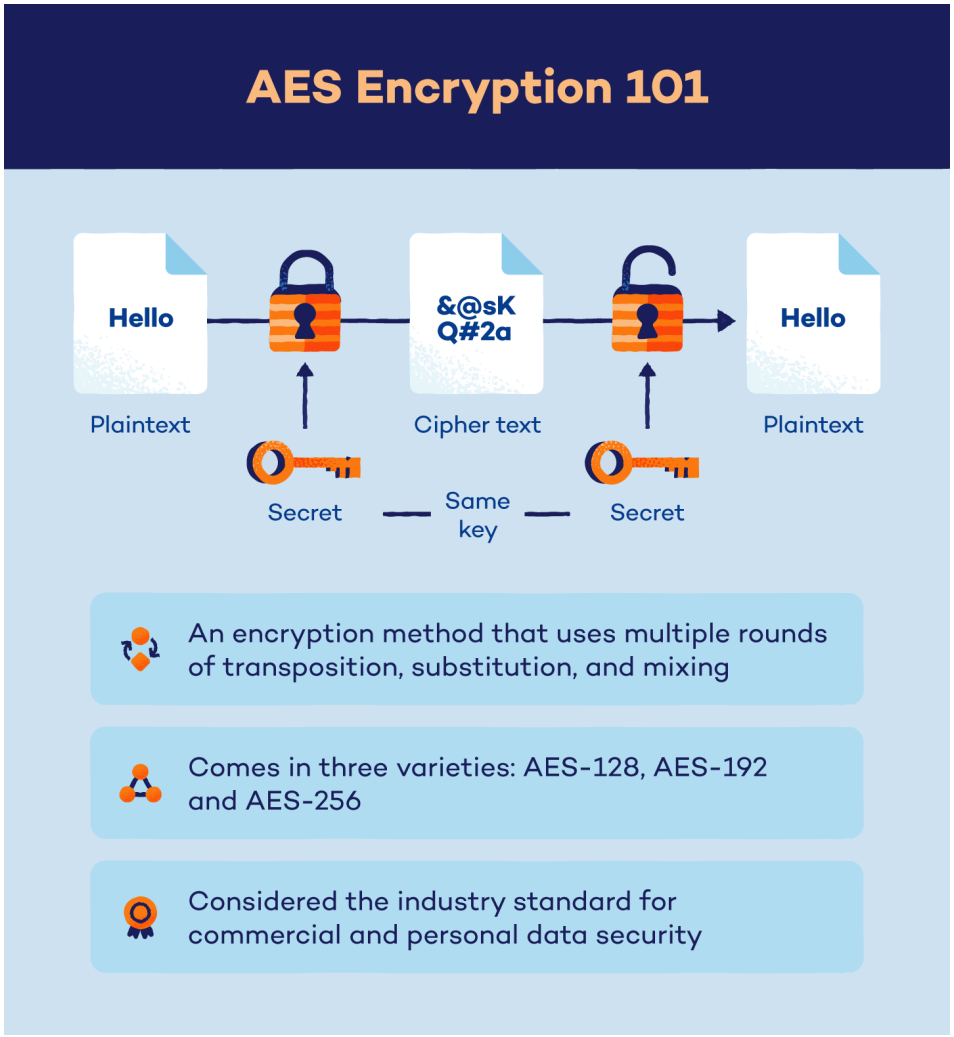
## How It Works

The AES encryption algorithm is a symmetric key encryption algorithm, meaning that the same key is used to encrypt and decrypt the data. The key is shared between the sender and the recipient of the encrypted data and is kept secret to prevent unauthorized access.

The strength of the AES encryption algorithm lies in its ability to resist attacks by brute force. Brute force attacks involve trying all possible combinations of keys until the correct key is found. With a key size of 128 bits, there are approximately 3.4 x 10^38 possible keys, making it practically impossible to break the encryption by brute force.

Another advantage of the AES encryption algorithm is its efficiency. It can encrypt and decrypt data quickly without too much processing power. This makes it suitable for applications requiring real-time encryption and decryption, such as online transactions and secure communication.

Example below :



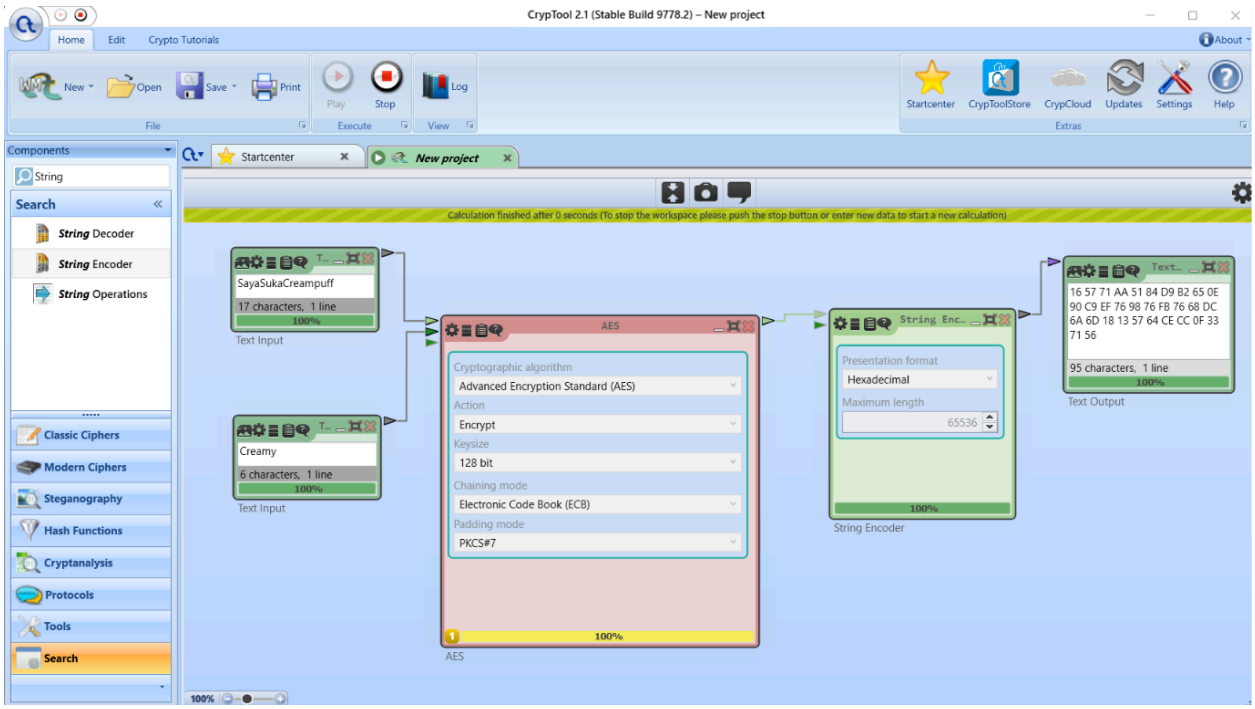
# Proof of Concept

Encryption a plain text into hexadecimal using [CrypTool 2](CrypTool 2).

In this encryption process in CrypTool 2, I used two(2) text inputs, one(1) AES modular block, one(1) string encoder as well as one(1) text output.

Plain text : **SayaSukaCreampuff**

SecretKey : **Creamy**




Encrypted cipher :

**16 57 71 AA 51 84 D9 B2 65 0E 90 C9 EF 76 98 76 FB 76 68 DC 6A 6D 18 13 57 64 CE CC 0F 33 71 56**

Justification :

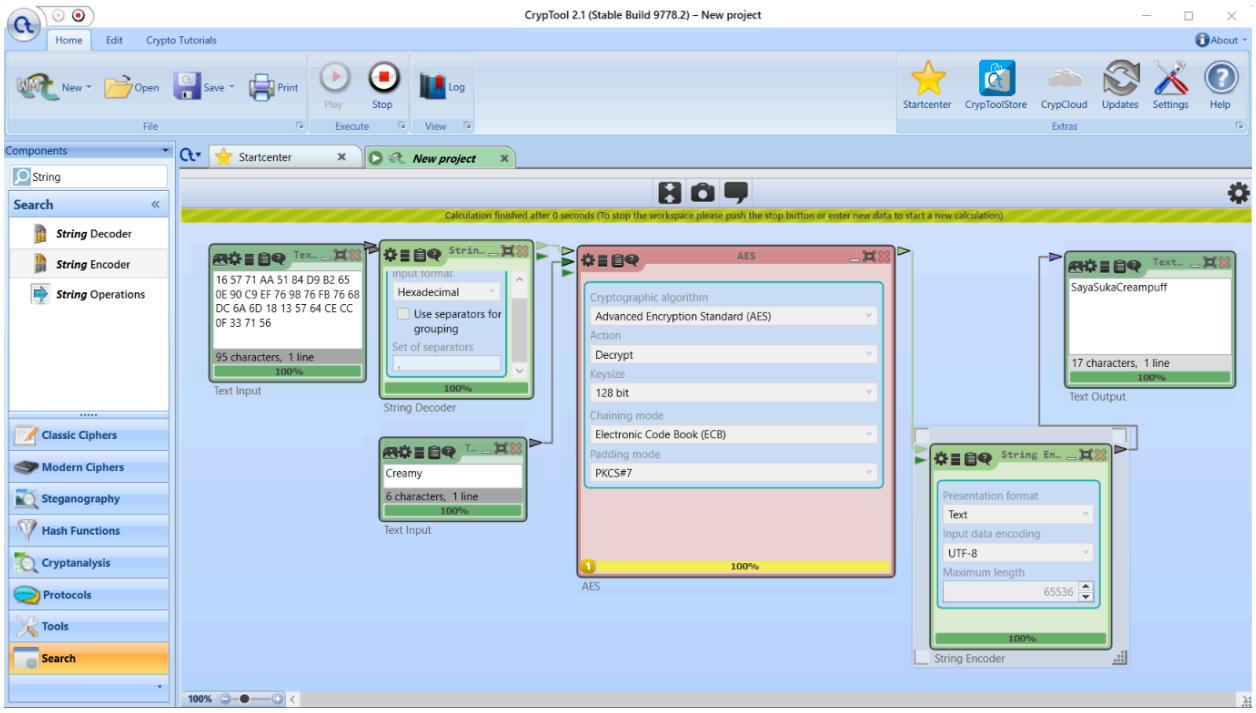
Converts human-readable plaintext (SayaSukaCreampuff) into unreadable hexadecimal ciphertext using a secret key (Creamy) via the AES cipher to prevent unauthorized access and secure data transmission.

Encryption a plain text into hexadecimal using CrypTool 2.

In this decryption process, I used two(2) text inputs, one(1) AES modular block, one(1) string decoder and one(1) string encoder as well as one(1) text output.

Cipher text : **16 57 71 AA 51 84 D9 B2 65 0E 90 C9 EF 76 98 76 FB 76 68 DC 6A 6D 18 13 57 64 CE CC 0F 33 71 56**

SecretKey : **Creamy**

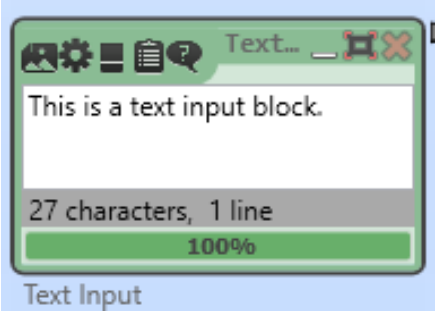



Decrypted text : **SayaSukaCreampuff**

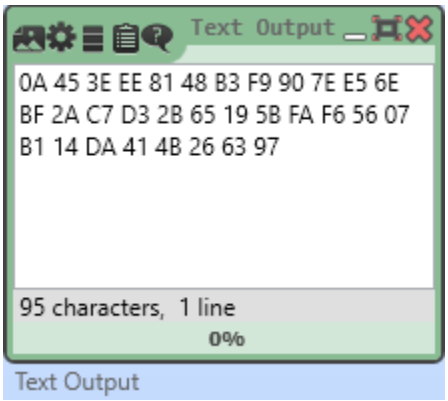
Justification :

Reverts unreadable hexadecimal ciphertext back into readable plaintext (SayaSukaCreampuff) using the identical secret key (Creamy) via the AES cipher to retrieve the original message.

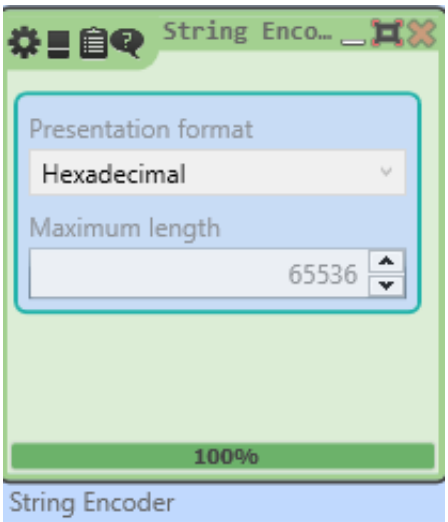
# Modular Blocks



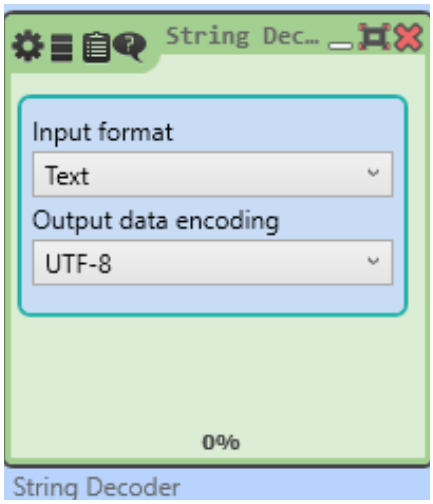

**Text Input**: Serves as the entry point for feeding raw string data, such as plaintext or secret key parameters directly into the workflow.



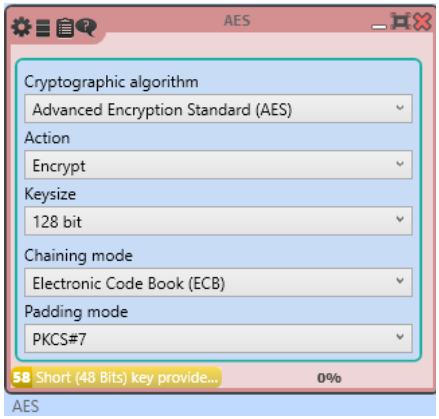

**Text Output:** Acts as the terminal node to capture and present the final processed results to the user.




**String Decoder**: Converts text strings formatted as hexadecimal into a standard byte stream required for input into cryptographic operations.




**String Encoder**: Reformats output byte streams into readable text representation formats (such as Hexadecimal).




**AES Cipher Block**: Executes the Advanced Encryption Standard mathematical algorithm using specified parameters (128-bit key, ECB mode, PKCS#7 padding) to encrypt or decrypt incoming byte streams.

# References


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